Domain Specialist

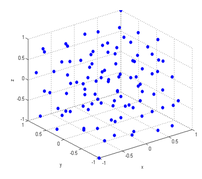
Parameter Bounds

Sample Size

Inverse Methods Expert

Research Software Engineer

Sampler



Geometry

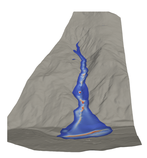
Inputs

Observation

Calibrated Parameter

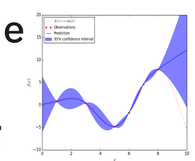
Initial Condition

Forward Model/ Simulator

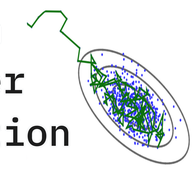


Design

Surrogate Model/ Emulator



Bayesian Parameter Calibration



Performance Indicator

Boundary Condition

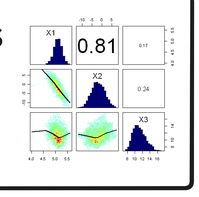
Model Parameter

Response

Outputs

Statistics

Diagnosis



Data Scientist

Computational Lab

Inputs

Geometry

BC/IC

Parameter

Sample

Outputs

Field

QoI

Visual

Report

Data

Input Data

Computational Unit

Model/ Algorithm

>>
10101010101
00100101000
01011000

Compute Environment

Software Ecosystem

Hardware Platform

Output Data

OpenFOAM

RobustGaSP

SALIB

emcee The MCMC Hammer

ArviZ

Models & Algorithms

Hardware Platform

CPU

GPU

Local

HPC

Cloud

Software Ecosystem

Compute Environment